Homework #1

1. Select one real-world application of image processing or computer vision. Research and analyze the technical roadmap of this application, then prepare a brief report answering the following questions (**15 pts**):

- Clearly describe the selected image processing or computer vision application. Explain the purpose and significance of this application.
- What type of images does the application acquire (e.g., medical images, satellite images, industrial images, etc.)? Describe the key image processing steps involved, such as: image pre-processing (filtering, noise reduction, enhancement), segmentation and object detection, feature extraction and classification
- Discuss the engineering and scientific requirements for this application. Consider aspects like image quality, computational speed, accuracy, reliability, and robustness. Identify any limitations or challenges associated with the current techniques used.

2. Suppose your smart phone has a 10-megapixel camera (1 megapixel = 10^6 pixels). Without any form of compression, how big would a 24bit-RGB color image be? Note that 24-bit RGB images are images in which each pixel is represented by three 8-bit quantities (thus, 24 bits total) specifying the intensities of red, green, and blue that form the color for the pixel. (Hint: 1 byte = 8 bits) **(5 pts)**

3. Which of the following are examples of electromagnetic (EM) waves? Check all that apply. **(5 pts)**

- ☐ Microwave
- ☐ Ripples in a lake
- ☐ Sound wave
- ☐ Light from the sun
- ☐ X-ray
- ☐ Ultrasound
- ☐ Infrared waves

4. For each of the systems described below, determine whether the system is (i) linear, and (ii)

shift invariant (**15 pts**):

(1) $g(x)=f(2x)=T[f(x)]$
(2) $g(m,n)=f(m-2,n+3)=T[f(m,n)]$
(3) $g(m,n)=f(m,n)\sin(m+n)=T[f(m,n)]$

5. You are provided with a sample grayscale image for testing various techniques for image processing in spatial domain. Your task is to perform several basic spatial-domain image processing operations and analyze both the processed images and their corresponding histograms. **(20 pts)**

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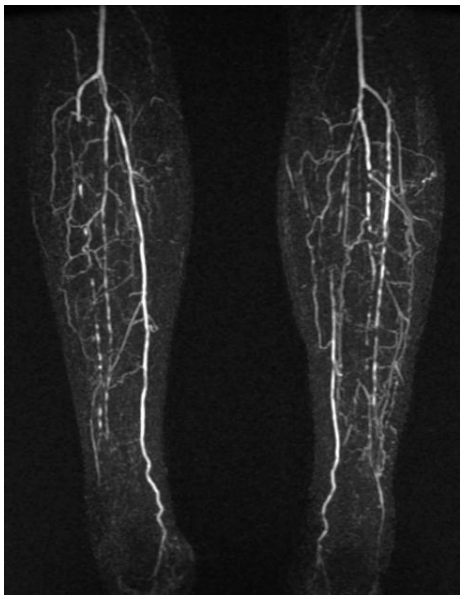


(a) Load the provided sample grayscale image into Python or Matlab. Display the original image and its histogram. Then add Gaussian noise to the image and display the noisy image together with its histogram. Briefly describe how the added noise affects the visual appearance and gray-level distribution of the image.

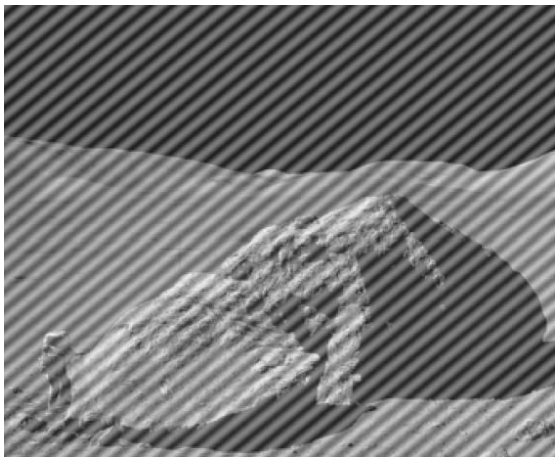
(b) Apply mean (averaging) filters with window sizes of 3×3, 5×5, and 9×9 to the noisy image. Display the filtered images and their corresponding histograms. Compare the results and explain how increasing the filter size affects noise reduction, image smoothness, edge sharpness, and fine detail preservation.

(c) Generate the negative of the original image using gray-level inversion. Display the negative image and its histogram.

(d) Apply gamma transformation to the original image using at least three different gamma values, including one value less than 1 and one value greater than 1. Display the transformed images and their histograms. Discuss how gamma values below and above 1 affect the brightness and contrast of the image, and explain the corresponding changes observed in the histogram.

(e) Perform intensity slicing on the original image by selecting a suitable gray-level range. You may either highlight the selected intensity range while suppressing the background or preserve the background while enhancing the selected intensities. Display the resulting image and its histogram. Explain what image features are emphasized by this operation and how the histogram reflects the selected intensity range.

6. You are provided with an image below corrupted by periodic noise, evident as regular patterns or stripes in the image. Your task is to use the 2D Fourier Transform (FFT) to filter out this periodic noise and restore the image. **(20 pts)**



(a) Image Analysis: Load the provided noisy image into MATLAB or Python.Compute and visualize the magnitude spectrum of the 2D FFT of the noisy image. Clearly identify and mark the frequency components responsible for the observed periodic noise.

(b) Noise Removal via Fourier Domain Filtering: Set these frequency components' magnitudes to the averaged magnitude value of the spectrum. Display and compare the magnitude spectrum before and after these operations.

(c) Restoration of the Image: Compute the inverse Fourier transform to restore the filtered image. Display the restored image and clearly explain the improvement compared to the original noisy image.

7. You are provided with 100 grayscale images of the same scene, each corrupted by independent Gaussian noise. Your task is to recover the image progressively by averaging multiple noisy observations, estimate the noise level from the averaged images, and study the scaling law of noise reduction. (**20 pts**)

(a) Load the provided 100 noisy grayscale images. Display several representative noisy images to illustrate the level of corruption. Then compute and display the mean images formed by averaging the first Nimages, where

$$N = 1,\ 2,\ 5,\ 10,\ 25,\ 50,\ 100$$

using

$$\bar{I}_N(x,y) = \frac{1}{N}\sum_{k=1}^{N} I_k\,(x,y)$$

Compare the visual quality of these mean images and briefly describe how the image appearance changes as the number of averaged images increases.

(b) Since the true noisy-free image is not available, estimate the noise level from a visually smooth region of the image that contains little edge or texture information. First use the mean image formed from all 100 images to select a suitable rectangular region of interest. Then use the same region of interest for all mean images. For each value of N, estimate the noise level as the standard deviation of the pixel intensities within that region, and report the corresponding estimated noise value. Plot the estimated noise level as a function of the number of averaged images N. To study the scaling law of noise reduction, plot the estimated noise level against Non log-log axes and fit the data to a power-law model of the form

$$\hat{\sigma}_N \approx CN^{-\alpha}$$

where $\hat{\sigma}_N$is the estimated noise level, C is a constant, and α is the fitted exponent. Determine the value of α from your data.

(c) Provide a theoretical derivation for the expected scaling law of noise reduction by image averaging. Assume that each noisy image can be written as

$$I_k(x,y) = f(x,y) + n_k(x,y)$$

where $f(x,y)$is the unknown true image and $n_k(x,y)$is independent zero-mean Gaussian noise with variance σ^2.